Cloud Notes

History of Cloud Computing

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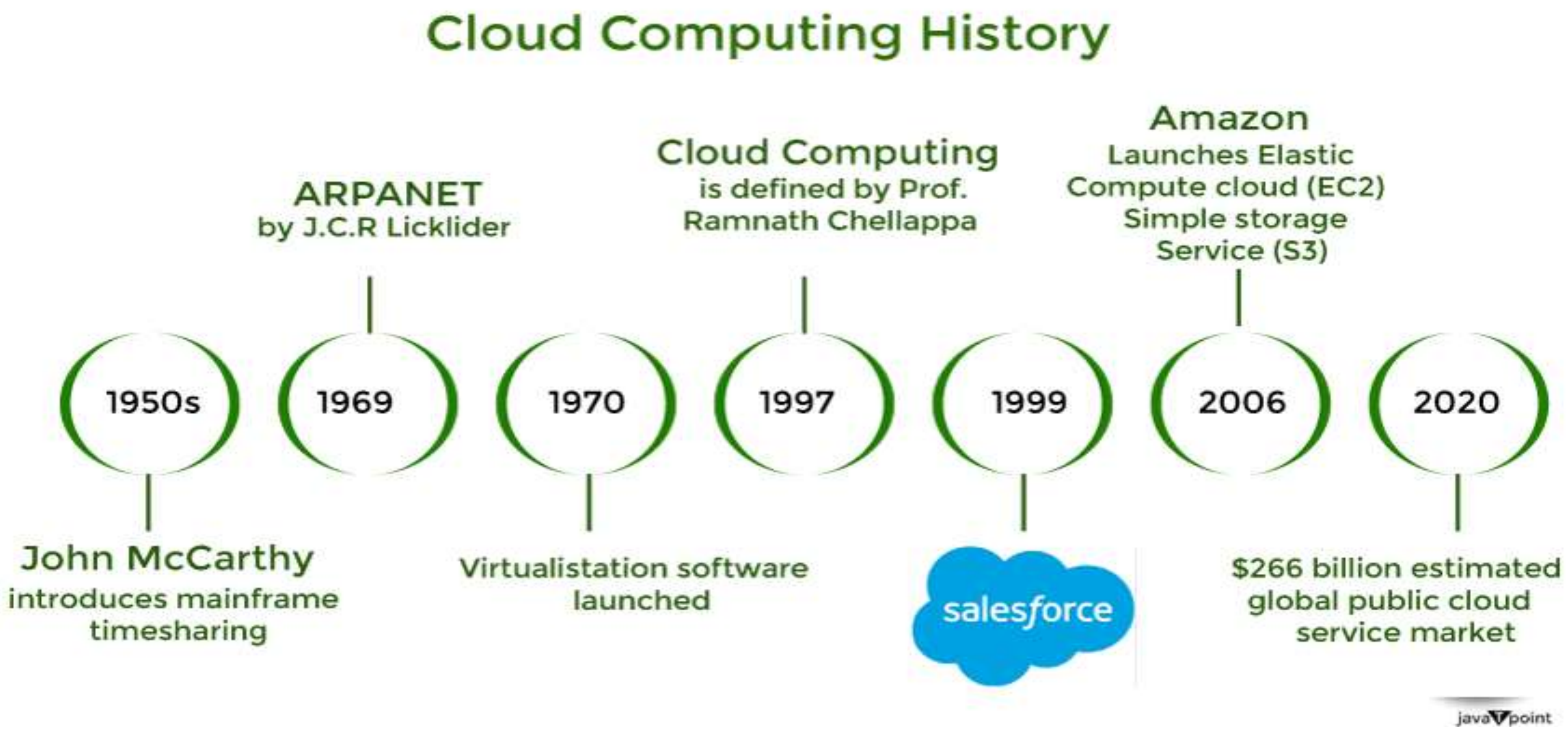


History of Cloud Computing

- Earlier Client Server Architecture was dominant. It has disadvantages related to resource sharing and scalability
- As the network grew the idea of Distributed computing evolved leading to parallel processing The centralized approach underwent a dramatic change with the advent of distributed computing, opening the door for more scalable and adaptable computer structures.
- Cloud Computing: The objective was to offer network-based, primarily the Internet, on-demand access to shared computer resources and services. The goal of cloud computing was to offer consumers flexible, scalable, and economical access to computing resources, storage, and applications. It shifted the emphasis to distant services and pay-as-you-go business models from local infrastructure and ownership.

History of Cloud Computing

- 1961, the idea of computers as a utility, comparable to water or electricity, was put out by Mr. John McCarthy at MIT.
- Delivering applications over the internet in 1999, Salesforce.com transformed the software sector.
- Launched in 2002, Amazon Web Services (AWS) first provided computing and storage services.
- Elastic Compute Cloud (EC2)'s launch in 2006 was what truly revolutionized cloud computing.
- Google Apps, which started offering cloud-based enterprise apps in 2009.
- Microsoft introduced Windows Azure, a feature-rich cloud computing platform, in the same year

Definition and Characteristics of Cloud

- Definition:

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. (NIST)

Definition and Characteristics of Cloud

- Characteristics

✓ **On-demand self-service**: A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.

✓ **Broad network access:** Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).

✓ **Resource pooling:** The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. There is a sense of location independence in that the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter). Examples of resources include storage, processing, memory, and network bandwidth.

✓ **Rapid elasticity:** Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.

✓ **Measured service:** Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

✓ **Multi-tenancy: I**n cloud computing, multitenancy means that multiple customers of a cloud vendor are using the same computing resources. Despite the fact that they share resources, cloud customers are not aware of each other, and their data is kept totally separate.

Benefits/Need of Cloud Computing

- Cost Efficient: Businesses can save costs on hardware, energy, and in-house technical specialists
- Faster deployment and market release : With cloud services, you can **deploy within days,** not months. Cloud computing allows you to **integrate and test new technologies** quickly.
- Ease of Scaling: Cloud’s flexible infrastructure allows you to scale on-demand.
- Remote Access and Availability: Cloud computing **offers remote access**.
- Collaboration and Productivity: Cloud computing enhances collaboration as workers **get access to data in real-time.**

Disadvantages of Cloud

- ✓risk of vendor lock-in
- ✓less control over underlying cloud infrastructure
- ✓concerns about security risks like data privacy and online threats
- ✓integration complexity with existing systems
- ✓unforeseen costs and unexpected expenses

Cloud Deployment Models

Four Cloud Deployment Models: Deployment models describe the ways with which the cloud services can be deployed or made available to its customers, depending on the organizational structure and the provisioning location. Four deployment models are usually distinguished, namely, public, private, community, and hybrid cloud service usage:

1. Private cloud: The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises. IBM Cloud, HPE Hewlett Packard
2. Public cloud: The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider. AWS, GCP, Azure

Cloud Deployment Models

- Community cloud: The cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be managed by the organizations or a third party and may exist on premise or off premise. AWS GovCloud

- Hybrid cloud: The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

Cloud Service Models

- **Three Service Offering Models**
  - The three kinds of services with which the cloud-based computing resources are available to end customers are as follows: Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS). It is also known as the software–platform–infrastructure (SPI) model of the cloud.
    - SaaS
    - PaaS
    - IaaS

Cloud Service Models

SaaS

- The capability provided to the consumer is to use the provider’s applications running on a cloud infrastructure, including network, servers, operating systems, storage, and even individual application capabilities, with the possible exception of limited user-specific application configuration settings.
- The applications are accessible from various client devices, either through a thin client interface, such as a web browser (e.g., web-based e-mail), or a program interface.
- The consumer does not manage or control the underlying cloud infrastructure.

Cloud Service Models

PaaS

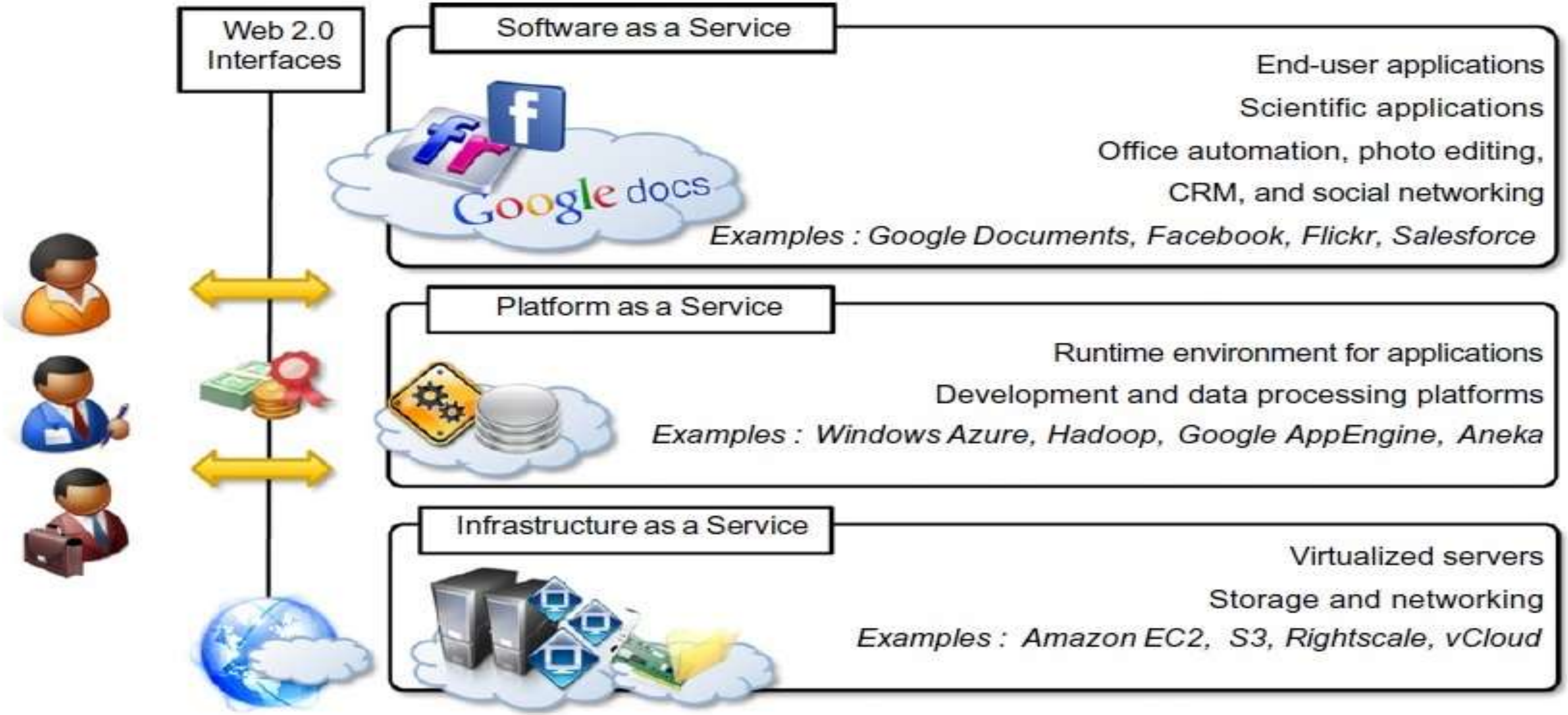
- The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider.
- The consumer does not manage or control the underlying cloud infrastructure, but has control over the deployed applications and possible configuration settings for the application-hosting environment.
- In other words, it is a packaged and ready-to-run development or operating framework.
- The PaaS vendor provides the networks, servers, storage and manages the levels of scalability and maintenance.

Cloud Service Models

IaaS

- The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources on a pay-per-use basis where he or she is able to deploy and run arbitrary software, which can include operating systems and applications.
- The consumer does not manage or control the underlying cloud infrastructure but has control over the operating systems, storage, and deployed applications and possibly limited control of select networking components (e.g., host firewalls).
- The service provider owns the equipment and is responsible for housing, cooling operation, and maintenance.

Cloud Service Models



Traditional IT vs. Cloud Services Models